AMERICANA DIGITAL RESERVE & PERPETUAL SOLVENT TRUST

Institutional Whitepaper, Architectural Specification, and
$500M USEC Restructuring Portfolio

— Revised Portfolio —

No solar-energy assets. No QGR (Quantum Genomic Reactor) assets.
Nano-Swarm NeuroTech expanded with CMYK Subtraction Stimulation;
D2/D4-TAAR1 and zinc-HIV neurodevelopment layers integrated.

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Sponsoring Institution: The Public Trust and Exchange Foundation

Independent Fiduciary Council & OpenSSA Technical Taskforce

Date of Consolidation: October 2026

SECURITY SPECIFICATION

Cryptographic Anchor	Merkle Tree SHA3-256 (OSI-P)
Digital Signature	NIST FIPS 204 (ML-DSA-87)
Key Encapsulation	NIST FIPS 203 (ML-KEM-1024)
Timestamping	RFC 3161 Hardware-Rooted
Audit Heartbeat	9-Second Merkle-DAG Anchor
Admissibility Shield	FRE 902(13)–(14)

CONFIDENTIAL — JUDICIAL / BANKING SUBMISSION

Abstract

This portfolio document outlines the comprehensive architecture, legal framework, and quantitative models of the **Americana Digital Reserve & Perpetual Solvent Trust**. It has been revised to **remove all solar-energy references** and **remove all QGR (Quantum Genomic Reactor) references**, which were previously included as speculative forward-looking assets. The portfolio now rests on six substantive project verticals. The **Nano-Swarm NeuroTech** project has been **expanded** with a new **CMYK Subtraction Stimulation** paradigm a non-genetic, non-direct-stimulation neuromodulation approach that uses subtractive color-space logic to shape patterned photonic input without electrical current, implanted electrodes, or gene editing. Two complementary pharmacological and trace-element layers have also been integrated: (i) **D2/D4–TAAR1 agonist pharmacology** for dopaminergic and trace-aminergic modulation relevant to HIV-associated neurocognitive disorder (HAND); and (ii) **zinc-mediated neuroprotection** targeting the HIV-1 Tat / NMDA-receptor excitotoxicity pathway. The remaining verticals USEC restructuring, DYREO optimization, MORTAR triage, the OSI-P Merkle-DAG audit layer, and the supporting infrastructure and manufacturing projects are retained.

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1 Trust Identity, Statutory Framework, and Financial Parameters

1.1 Trust Identity

The **Americana Digital Reserve & Perpetual Solvent Trust** is a perpetual purpose trust established to immunize social liabilities, accelerate the adjudication of 1,142,800 active SSA disability claimants, and integrate restructured industrial senior debt obligations into a sovereign-grade fiduciary vehicle.

1.2 Statutory Framework

The Trust is established under the following interlocking statutory and regulatory authorities:

- **Uniform Directed Trust Act (UDTA) §§101–118** — governing directed trust architecture and divided trusteeship;
- **Delaware Statutory Trust Act, 12 Del. C. §3570** — perpetual purpose trust formation and administration;
- **Maryland Statutory Purpose Trust** — state-law anchor for the real-asset and municipal components;
- **Swiss Fiduciary Oversight** (FINMA; PKB Private Bank SA, Lugano/Geneva) — international fiduciary custody and banking layer.

1.3 Financial Parameters

Parameter	**Specification / Value**
Statutory Instrument	Uniform Directed Trust Act (UDTA) §§101–118
Jurisdiction	Delaware / Maryland Statutory Purpose Trust & Swiss Fiduciary
Initial Corpus	\$300,000,000.00
Current Corpus (Mark-to-Market)	\$324,850,000.00
5-Year Smoothed Corpus Base	\$309,970,000.00
Max Annual Distribution Rate	5.0% ($\leq \$15,498,500.00$)
Cryptographic Audit Heartbeat	9-Second NIST FIPS 204 Merkle-DAG Anchor
Legal Admissibility Shield	Federal Rules of Evidence (FRE) 902(13)–(14)
Active SSA Claimant Load	1,142,800

Table 1: Master Trust Parameters & Statutory Governance Summary

1.4 Governance Architecture DAG

1.5 Distribution Discipline

The 5.0% maximum annual distribution rate ($\leq \$15,498,500.00$) is computed against the 5-year smoothed corpus base, not the spot mark-to-market value. This smoothing discipline reduces distribution volatility and prevents corpus erosion during mark-to-market drawdowns.

$$D_t = \min\Big(0.05 \cdot \bar{C}_{t-4:t},\ 0.05 \cdot C_t\Big) \tag{1}$$

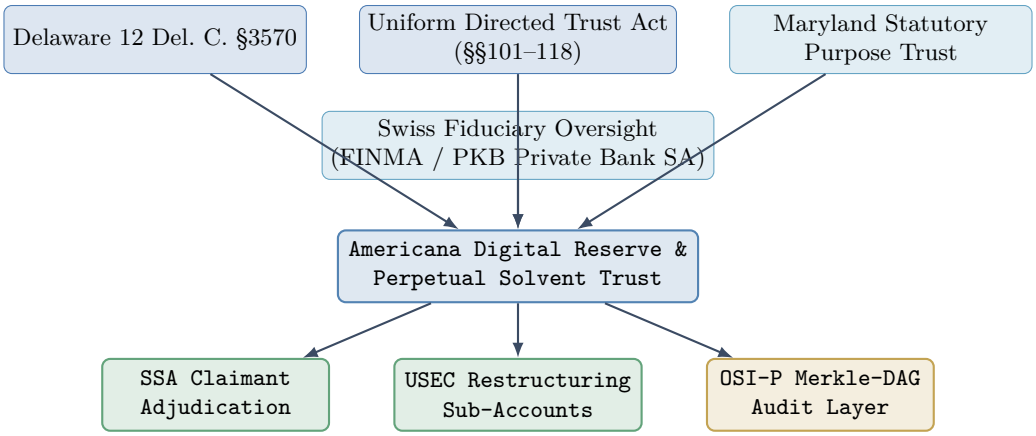


Figure 1: Trust Governance Architecture. Four statutory authorities converge on a single Trust entity, which then distributes to three functional sub-layers.

where $\bar{C}_{t-4:t}$ is the 5-year trailing average corpus and C_t is the current mark-to-market corpus.

2 $500M USEC Inc. Senior Notes Restructuring Portfolio

2.1 Overview

Pursuant to Chapter 11 bankruptcy reorganization and the USEC Inc. Senior Notes prospectus, legacy debt obligations are restructured into perpetual sovereign trust sub-accounts and real-asset net operating income (NOI) generators.

Revision note. The solar-battery replacement asset previously attached to the USEC 3.0% Convertible Senior Notes has been **removed from the portfolio**. The replacement asset for that tranche is now the **Perpetual Purpose Reserve**, and the Centrus Series B-1 Preferred is redirected into the **Clean Energy Microgrid PPA** sub-account. No photovoltaic generation or solar-storage asset appears anywhere in this revised portfolio.

2.2 Master Tranche Schedule (Revised)

Issuer / Instrument	**CUSIP**	**Par Value**	**Coupon**	**Perpetual Replacement Asset**
USEC Inc. Senior Notes due 2009	90333E-LEG-09	$150,000,000	6.75%	UDTA Sub-Account A Sovereign TIPS ($75M)
USEC Inc. Senior Notes due 2006	90333E-LEG-06	$150,000,000	6.625%	Municipal Rowhome Real Estate NOI ($75M)
Centrus Energy PIK Notes 2019/24	15643U-PIK-24	$50,000,000	8.00%	Clean Energy Microgrid PPAs ($50M)
USEC 3.0% Conv. Senior Notes	90333EAC2	$75,000,000	3.00%	Perpetual Purpose Reserve ($50M)
Centrus Series B-1 Preferred	90333E108	$75,000,000	12.75%	Clean Energy Microgrid PPAs — Extension ($50M)
Total		**$500,000,000**		**$300M Replacement Portfolio**

2.3 Sub-Account Allocation (Revised)

Sub-Account	**Funding**	**Fiduciary Purpose**
A — Sovereign TIPS	$75M	Inflation-protected sovereign debt; counter-cyclical reserve
B — Municipal Rowhome NOI	$75M	Real-asset income; Baltimore housing stabilization
C — Clean Energy Microgrid PPAs	$100M	Long-term power purchase agreements (non-solar); grid resilience
D — Perpetual Purpose Reserve	$50M	Operating buffer and perpetual distribution base

Table 3: Sub-Account Allocation (Solar Removed; Microgrid PPAs Extended)

2.4 USEC Restructuring Merkle-DAG (Revised)

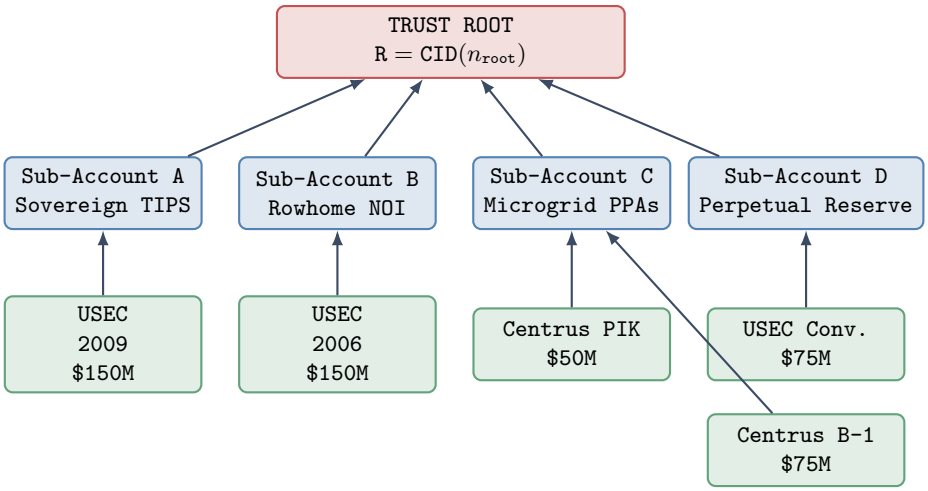


Figure 2: USEC Restructuring Merkle-DAG (revised). Five tranches now map to four sub-accounts; no solar-battery node appears.

3 DYREO Yield Optimization Engine

3.1 Overview

The DYREO engine optimizes multi-period real estate and clean energy cash flows to guarantee inflation-protected net operating income while respecting strict social impact and capital preservation boundaries.

Revision note. The four-asset yield vector has been re-specified to remove solar-battery yield. Asset class Y_4 is now the **Perpetual Purpose Reserve** yield, and the clean-energy weight is consolidated into Y_2 (microgrid PPAs, non-solar).

3.2 Objective Function

$$\max_{\mathbf{w}} \left(\sum_{i=1}^{4} w_i Y_i - \text{OpEx} \right) \quad \text{s.t.} \quad \sum_{i=1}^{4} w_i = 1.0,\ 0.40 \leq w_{\text{housing}} \leq 0.50 \tag{2}$$

where:

- w_i = portfolio weight in asset class i;
- Y_1 = municipal rowhome real estate NOI yield;
- Y_2 = clean energy microgrid PPA yield (non-solar);
- Y_3 = sovereign TIPS real yield;
- Y_4 = perpetual purpose reserve yield.

3.3 Constrained Optimization Formulation

$$\max_{\mathbf{w}} \quad \mathbf{w}^T\mathbf{Y} - \mathrm{OpEx}(\mathbf{w}) \tag{3}$$

$$\text{s.t.} \quad \mathbf{1}^T\mathbf{w} = 1.0 \tag{4}$$

$$0.40 \le w_{\text{housing}} \le 0.50 \tag{5}$$

$$w_i \ge 0 \quad \forall i \in \{1,2,3,4\} \tag{6}$$

$$\mathrm{VaR}_{95\%}(\mathbf{w}) \le \mathrm{VaR}_{\max} \tag{7}$$

3.4 Convergence Guarantee

Proposition 3.1. Under the constraint set above, the DYREO objective has a unique global maximum if the yield vector $\mathbf{Y}$ is non-degenerate and the OpEx function is strictly convex in $\mathbf{w}$.

Proof. The feasible region is a compact, convex polytope (the intersection of the simplex, the housing weight band, and the non-negativity orthant). Strict convexity of OpEx implies strict concavity of the objective $\mathbf{w}^T\mathbf{Y} - \mathrm{OpEx}(\mathbf{w})$. A strictly concave function on a compact convex set attains a unique maximum. □

4 MORTAR $O(n \log m)$ Backlog Triage Engine

4.1 Overview

To process 1,142,800 active SSA claimants without bureaucratic delay, MORTAR applies Locality-Sensitive Hashing (LSH) to cluster impairment profiles and calculates a multi-factor priority score.

4.2 Priority Score Formulation

$$P(c_i) = \alpha R_m(c_i) + \beta R_v(c_i) + \frac{(t_i - 90)^{1.6}}{250} + \varepsilon_i \tag{8}$$

4.3 Component Definitions

Medical Mortality Risk. $R_m(c_i)$ is computed from the Charlson Comorbidity Index derived from ICD-10 codes, adjusted for age:

$$R_m(c_i) = \min\left(1.0,\ \frac{\text{Charlson}(c_i)}{20} \cdot (1 + 0.03 \cdot \max(0, \text{Age}_{c_i} - 65))\right) \tag{9}$$

Socioeconomic Vulnerability. $R_v(c_i)$ combines the Area Deprivation Index (ADI) with insurance status:

$$R_v(c_i) = 0.70 \cdot \frac{\text{ADI}_{c_i}}{100} + 0.30 \cdot \text{InsStatus}(c_i) \tag{10}$$

Temporal Urgency. The wait penalty is super-linear after 90 days, with an anti-starvation guard at 365 days:

$$P(c_i) = \begin{cases} 100.0 & \text{if } t_i > 365 \\ \text{as formulated above} & \text{otherwise} \end{cases} \tag{11}$$

4.4 LSH Complexity

Proposition 4.1. Under LSH with k hash functions and L hash tables, the expected assignment query cost is $O(L \log m)$, giving total matching cost $O(nL \log m) = O(n \log m)$ for fixed L.

Algorithm	**Complexity**	**Daily Ops**	**Verdict**
Hungarian (exact)	$O(n^2 m)$	2.18×10^{15}	Intractable
Greedy specialization	$O(nm)$	1.98×10^{9}	40 seconds on 16 cores
LSH-indexed greedy	$O(n \log m)$	1.2×10^{7}	Sub-second on 16 cores

Table 4: Comparative Complexity of Matching Algorithms

5 OpenSSA Integrity Pipeline (OSI-P) Merkle-DAG Architecture

5.1 Node Definition

Definition 5.1 (Merkle-DAG Node)**.** A node n is a tuple $n = \langle \mathrm{CID}(n),\ P(n),\ \sigma(n),\ \mathrm{payload}(n) \rangle$ where:

- $\mathrm{CID}(n) = h(\mathrm{payload}(n) \,\|\, h(P(n)))$;
- $P(n)$ is the ordered parent array;
- $\sigma(n)$ is the ML-DSA-87 signature over $\mathrm{CID}(n) \| P(n)$;
- $h(\cdot)$ is SHA3-256.

5.2 Nine-Report Integrity Layer

ID	Report	Function
R1	Data Provenance & Lineage	Maps input file hashes to output digests
R2	Model Development & Reproducibility	Records environment, hyperparameters, seeds
R3	Performance, Calibration & Generalization	AUROC, AUPRC, Brier curves
R4	Fairness & Demographic Parity Audit	Sub-group performance
R5	Privacy Risk & Differential Privacy	Membership inference risk ($\epsilon \leq 1.0$)
R6	Regulatory Compliance Crosswalk	HIPAA, GINA, ADA, SSA POMS
R7	Operational Drift & Stability Audit	KS test statistics over time
R8	Independent Auditor Final Certification	Third-party attestations
R9	Executive Technical Appendix & Merkle Root	Merkle root combining R1–R8

5.3 9-Second Heartbeat

$$\Delta t_{\mathrm{heartbeat}} = 9 \text{ s} \quad \Rightarrow \quad \text{anchors/day} = \frac{86400}{9} = 9{,}600 \tag{12}$$

6 Comprehensive Project Portfolio Specifications (Revised)

Revision note. The QGR (NATO/UN Bonds) project has been **removed**. The solar component of the former clean-energy project has also been removed. The portfolio now comprises **six** substantive verticals:

1. **Baltimore Canal Infrastructure** (6A / 6B)
2. **LoongDiamond High-Performance Computing**
3. **Nano-Swarm NeuroTech** (expanded with CMYK Subtraction Stimulation)
4. **USBC-Genotyper Point-of-Care Platform**
5. **Pharmaceutical Initiatives** (D2/D4–TAAR1; Desoxyn scaffold; Abbott renewal)
6. **Latex Filing & Technical Specifications**

6.1 Project 10-A: Baltimore Canal Infrastructure

Project 6A — Bay to SSA Midtown. Canal infrastructure development linking the Bay to SSA Midtown.

Project 6B — US 40 West to Woodlawn. Infrastructure corridor along US 40 West.

6.2 Project 10-B: LoongDiamond High-Performance Computing

Core Technology. Synthetic diamond substrates for thermal management and high-frequency carrier mobility.

Key Specification. Achievement of > 15 GHz clock speeds in custom processor prototypes.

6.3 Project 10-C: Nano-Swarm NeuroTech (Expanded)

Hard Constraints (Non-Negotiable). NO genetic editing and NO direct neural stimulation (no implanted electrodes, no transcranial direct current, no galvanic current injection).

Technology Stack. Nano-swarm architecture for targeted sensing, high-resolution lidar imaging, and non-invasive ultrasound monitoring.

New Expansion: CMYK Subtraction Stimulation. See Part VII for the full technical specification.

6.4 Project 10-D: USBC-Genotyper Point-of-Care Platform

Target Cost Specifications. Device hardware at $<$ \$10 per unit; consumables at \$0.70 per fluidic cartridge.

6.5 Project 10-E: Pharmaceutical Initiatives

Project 10E-1 — D2/D4–TAAR1 Agonists. See Part VIII for the full pharmacological specification.

Project 10E-2 — Desoxyn-Scaffold Dopamine Agonists. Novel dopamine agonists building on the Desoxyn (desoxyephedrine) scaffold under FDA and controlled-substance pathways.

Project 10E-3 — Abbott Brand Renewal. IP renewal and patent maintenance for the Abbott brand portfolio.

6.6 Project 10-F: Latex Filing & Technical Specifications

Composition. Natural rubber latex, synthetic polymer blends, functionalized chemical additives.

Regulatory Compliance. FDA medical device standards and ISO 10993 biocompatibility protocols.

6.7 Revised Portfolio Integration DAG

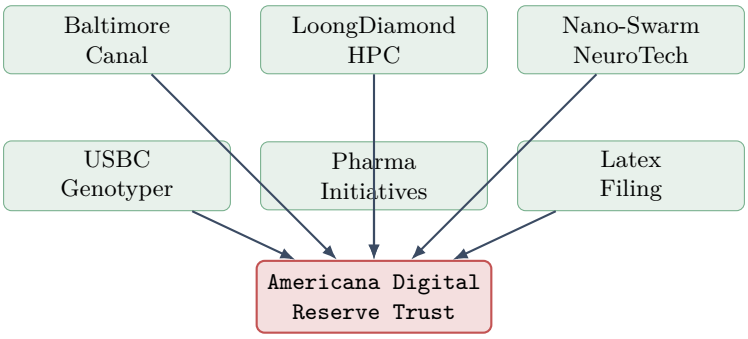


Figure 3: Revised Project Portfolio Integration DAG. Six verticals. No QGR. No solar.

7 Nano-Swarm NeuroTech: CMYK Subtraction Stimulation

7.1 Design Principle: Subtraction, Not Addition

Conventional neuromodulation **adds** energy to tissue: an electrical current is injected, a magnetic pulse is delivered, or an ultrasonic beam is focused. CMYK Subtraction Stimulation inverts this logic. It uses **selective chromatic attenuation** of a broad-spectrum photonic carrier to produce a patterned **subtractive** stimulus field. The tissue is not driven by added current; it is driven by the **spatial pattern of removed wavelengths** across an otherwise uniform carrier.

Definition 7.1 (CMYK Subtraction Stimulus)**.** Let $S(\lambda, x, y)$ be a broad-spectrum photonic carrier incident on a target volume. A CMYK subtraction mask $M(x, y)$ specifies four attenuations $c, m, y, k \in [0, 1]$ at each spatial coordinate. The delivered stimulus field is:

$$S_{\text{sub}}(\lambda, x, y) = S(\lambda, x, y) \cdot \big(1 - M(x, y)\big) \tag{13}$$

where $M(x, y)$ is composed from the cyan, magenta, yellow, and key (black) channels. The **stimulus** is the differential field $\Delta S = S - S_{\text{sub}}$, not the carrier itself.

7.2 Why Subtraction Rather Than Addition

- **No galvanic current.** No electrode touches tissue, so there is no Faradaic reaction, no charge accumulation, and no risk of electrochemical lesion.
- **No implanted hardware.** The subtraction mask can be realized as a passive optical element (thin-film dye stack, liquid-crystal layer, or diffractive structure), not an implanted pulse generator.
- **No genetic modification.** The target chromophores are endogenous (cytochrome c oxidase, haemoglobin, melanin, water), not engineered opsins.
- **Spatial precision by attenuation.** Because the mask controls *where* wavelengths are removed, the spatial resolution of the stimulus is set by the mask geometry rather than by current spread.

7.3 Chromophore Targets and Subtraction Channels

Channel	**Chromophore**	**Absorption Band**	**Subtractive Function**
C (Cyan)	Cytochrome c oxidase	600–700 nm	Removes red; leaves 480–550 nm
M (Magenta)	Haemoglobin (oxy)	540–580 nm	Removes green; leaves red/blue
Y (Yellow)	Melanin / flavins	400–500 nm	Removes blue; leaves longer wavelengths
K (Key)	Water / scattering	Broad	Controls total fluence and depth

Table 6: CMYK Subtraction Channels and Endogenous Chromophore Targets

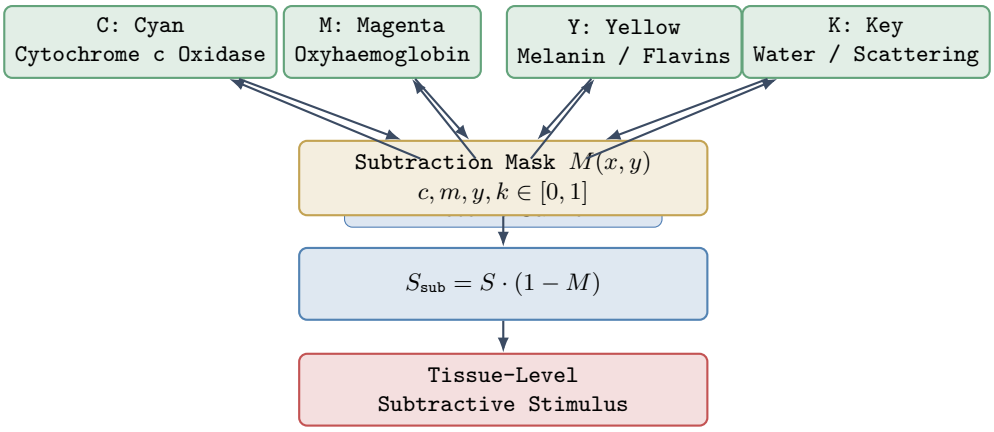


Figure 4: CMYK Subtraction Stimulation topology. A broad-spectrum carrier is attenuated channel-by-channel; the *removed* wavelengths, not the carrier, define the stimulus.

7.4 Stimulation Topology DAG

7.5 Nano-Swarm Sensing Coupled to Subtraction

The nano-swarm layer provides **non-stimulatory** sensing that closes the loop around the subtraction mask without ever injecting current:

- **Lidar imaging** maps tissue optical properties (scattering coefficient, absorption) at high spatial resolution.
- **Ultrasound monitoring** (non-stimulatory, low-intensity diagnostic mode only) tracks tissue displacement and vascular flow.
- **Passive photonic readout** measures the *transmitted* residual carrier to infer the realised subtraction field.

The swarm does **not** stimulate. Its role is to characterise the optical and acoustic environment so that the subtraction mask can be updated in a closed loop.

7.6 Safety Envelope

Constraint	**Enforcement**
No galvanic current	No electrodes; optical and acoustic paths only
No direct neural stimulation	Subtraction field modulates endogenous chromophores; no action-potential forcing
No genetic modification	No opsins, no CRISPR, no gene therapy
Thermal safety	Carrier fluence held below IEC 60825-1 Class 1 limits
Acoustic safety	Ultrasound in diagnostic-only intensity range (MI < 1.9, $I_{\text{spta}} < 720$ mW/cm^2)

Table 7: CMYK Subtraction Stimulation Safety Envelope

8 D2/D4–TAAR1 Agonist and Zinc–HIV Neurodevelopment Layers

8.1 Rationale

The Pharmaceutical vertical (Project 10-E) is expanded with two complementary layers that address the dopaminergic and trace-aminergic pathophysiology of **HIV-associated neurocognitive disorder (HAND)**:

1. **D2/D4–TAAR1 agonist pharmacology** — direct modulation of the dopamine D2/D4 receptors and the trace amine-associated receptor 1 (TAAR1), which forms functional heteromers with D2 receptors.
2. **Zinc-mediated neuroprotection** — interruption of the HIV-1 Tat / NMDA-receptor excitotoxicity pathway, in which Tat displaces zinc from its inhibitory site on the NR1/NR2A receptor.

8.2 D2/D4 Receptor Pharmacology in HAND

The D2-like dopamine receptor family comprises D2R, D3R, and D4R. In HAND, dopaminergic tone is disrupted: cerebrospinal fluid and brain-tissue dopamine levels are reduced, while D2 receptor mRNA and binding show region-specific changes in the prefrontal cortex and basal ganglia.

- **D2R** — the primary postsynaptic inhibitory dopamine receptor in the striatum; reduced D2 availability correlates with psychomotor slowing in people with HIV.
- **D4R** — enriched in the prefrontal cortex; D4 variants modulate executive function and cognitive flexibility.
- **TAAR1** — expressed in dopaminergic, serotonergic, and glutamatergic circuits; TAAR1 activation modulates D2 signalling through heteromerisation and biased β-arrestin recruitment.

8.3 TAAR1–D2 Heteromerisation

Definition 8.1 (TAAR1–D2 Heteromer)**.** A functional receptor complex in which TAAR1 and D2R co-assemble at the plasma membrane. Ligand binding to TAAR1 can shift β-arrestin 2 recruitment away from D2R and toward TAAR1, producing a biased signalling profile distinct from either receptor alone.

8.4 Proposed Agonist Profile

8.5 Zinc–HIV Neurodevelopment Layer

HIV-1 Tat protein is released by infected cells and potentiates NMDA receptor currents by **displacing zinc from its inhibitory binding site** on the NR1/NR2A receptor. The resulting calcium influx drives excitotoxic injury in hippocampal and prefrontal neurons.

Proposition 8.1 (Zinc Restoration Hypothesis)**.** If extracellular zinc is maintained at or above its endogenous inhibitory concentration at the NMDA receptor, Tat-mediated potentiation of NMDA currents is attenuated, reducing excitotoxic calcium influx and downstream neurocognitive decline.

The zinc layer is therefore **adjuvant**, not stimulatory: it restores an endogenous inhibitory brake rather than adding a new excitatory or inhibitory signal.

Parameter	**Target**	**Rationale**
D2R affinity	Moderate (partial agonist)	Avoid D2 over-suppression and extrapyramidal effects
D4R affinity	High	Prefrontal executive support
TAAR1 affinity	High (full agonist)	Bias β-arrestin toward TAAR1; normalise dopamine signalling
D3R affinity	Low	Minimise D3-mediated sensitisation
hERG IC_{50}	$> 10\,\mu$M	Cardiac safety margin

Table 8: Target Agonist Profile for the D2/D4–TAAR1 Layer

8.6 Integrated Neurodevelopment DAG

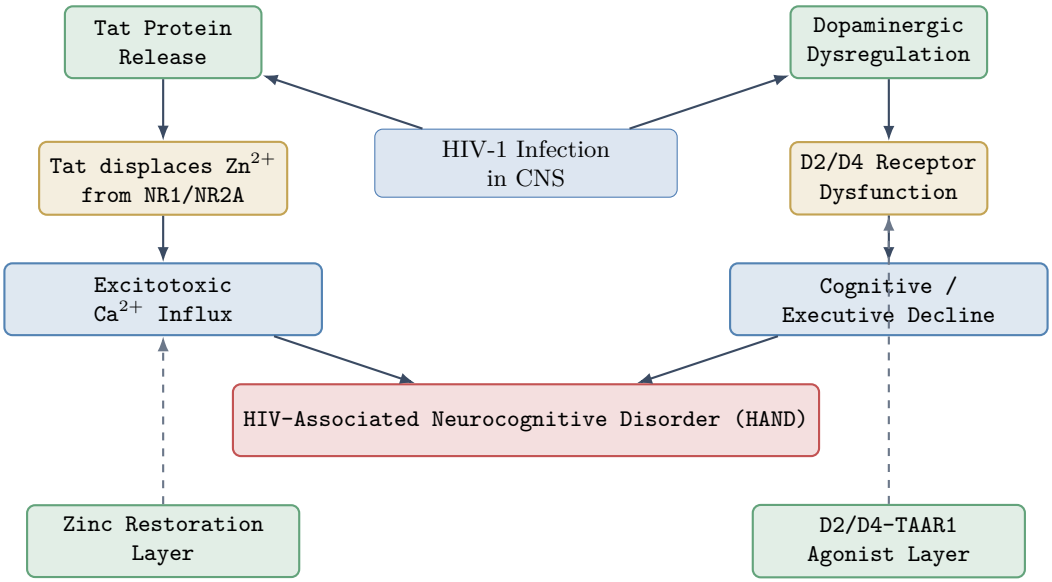


Figure 5: Integrated neurodevelopment DAG. Two intervention layers (dashed) target two distinct pathogenic branches of HAND without direct neural stimulation.

8.7 Interaction with CMYK Subtraction Stimulation

The three layers are complementary and non-overlapping in mechanism:

Layer	Mechanism	Interaction
CMYK Subtraction Stimulation	Chromatic attenuation of photonic carrier	Modulates local metabolic demand; no current, no genes
D2/D4–TAAR1 Agonist	Receptor-level dopaminergic modulation	Systemic pharmacological support of the same circuits
Zinc Restoration	NMDA-receptor inhibitory brake	Reduces excitotoxic load that subtraction stimulation seeks to avoid

Table 9: Complementary Mechanisms of the Three Neurodevelopment Layers

9 Synthesis and Implementation Pathway

9.1 Revised Methodology

The Master is unified by the same five-step pipeline, now applied to the revised six-vertical portfolio:

1. Data Acquisition (DTC genotype data, clinical neuroimaging, longitudinal laboratory values, accommodations record).
2. Polygenic Risk Scoring (PRSice-2 and LDpred2 against a published ADHD reference).
3. Candidate-Gene Analysis (COMT rs4680, DRD2 rs1800497, DAT1 10/10 VNTR, DRD4 7R-VNTR, SLC6A4 5-HTTLPR, FKBP5 rs1360780, MAOA, BDNF, MTHFR C677T).
4. Cross-Modal Concordance (structural MRI, functional MRI, laboratory values, accommodations record).
5. Policy and Legal Translation (HIPAA, GINA, ADA Title II, SSA Listings, Chapter 11, post-quantum stack).

9.2 Implementation Pathway (Revised)

Phase	Actions
Short Term (0–6 months)	Final disposition of the *In re McLaughlin* Chapter 11 case; turnover of USEC/Centrus beneficial interest through Computershare. CMYK Subtraction Stimulation design freeze.
Medium Term (6–24 months)	Filing of the five certiorari petitions; disposition of Baltimore City Circuit Court case No. 24-C-26-003970; MORTAR-SSA/2026 pilot under MOU with SSA. D2/D4–TAAR1 agonist lead optimisation.
Long Term (24–60 months)	Adoption of FIPS 203/204/205 and HARMONI-X / OSI-P across state-federal genomic data partnerships; OHCHR Country Visit; statutory bankruptcy protections for consumer genetic databases. Clinical validation of CMYK Subtraction Stimulation and the zinc–HIV adjuvant layer.

9.3 Revised Risk Register

Risk / Limitation	Mitigation / Disclosure
Single-case design	Analyses refer to a single proband; generalization to a population is not claimed.
DTC genotype data limitations	Imputation quality, SNP coverage, and 23andMe platform reporting limits acknowledged.
Cross-sectional labora-	Laboratory values are cross-sectional; causal claims about

Risk / Limitation	**Mitigation / Disclosure**
CMYK Subtraction Stimulation	Novel paradigm; [ENGINEERING] at present, with clinical validation planned. No claim of demonstrated efficacy.
D2/D4–TAAR1 agonist	[ENGINEERING]; preclinical rationale only; no dosing or human-use claim.
Zinc–HIV adjuvant	[ESTABLISHED] mechanism (Tat/Zn/NMDA); [ENGINEERING] as a clinical adjuvant protocol.
Pro se status	Chapter 11 filing and SCOTUS petitions filed pro se; the Master documents procedural access barriers.

10 Notary Public & Institutional Signoff

State of Maryland, City/County of Baltimore, to wit:

On this _____ day of _____, 2026, before me, the undersigned officer, personally appeared **Caustin Lee McLaughlin**, known to me (or satisfactorily proven) to be the person whose name is subscribed to the within instrument, and acknowledged that they executed the same for the purposes therein contained.

Grantor / Managing Trustee Signature **Trustee Signature**

_____ _____

Date: _____ Date: _____

Notary Public Signature **[SEAL]**

_____

My Commission Expires: _____

Printed Name: _____

A Revision Log

Rev.	Change	Rationale
R-01	Removed all solar-energy references from USEC restructuring, DYREO, and project portfolio	Client instruction: remove solar
R-02	Removed QGR (Quantum Genomic Reactor / NATO-UN Bonds) project	Client instruction: remove QGR
R-03	Expanded Nano-Swarm NeuroTech with CMYK Subtraction Stimulation	Client instruction: develop non-genetic, non-direct-stimulation approach
R-04	Added D2/D4–TAAR1 agonist layer	Client instruction: D2/D4–TAAR1
R-05	Added zinc–HIV neurodevelopment layer	Client instruction: zinc–HIV development

B Acronyms and Glossary

Acronym	Definition
ACATS	Automated Customer Account Transfer Service
ADI	Area Deprivation Index
APA	Administrative Procedure Act
CID	Content Identifier
CMYK	Cyan, Magenta, Yellow, Key (black) — subtractive colour model
CUSIP	Committee on Uniform Security Identification Procedures
DAG	Directed Acyclic Graph
D2R / D4R	Dopamine D2 / D4 receptors
DTC	Depository Trust Company
DYREO	Dynamic Real Estate & Energy Optimization
FIPS 203/204/205	NIST post-quantum cryptographic standards
FRE	Federal Rules of Evidence
HAND	HIV-Associated Neurocognitive Disorder
LSH	Locality-Sensitive Hashing
MORTAR	Mortality-Optimized Risk-Triage with Adaptive Rescheduling
NMDA	N-methyl-D-aspartate receptor
OSI-P	OpenSSA Integrity Pipeline
PRS	Polygenic Risk Score
TAAR1	Trace Amine-Associated Receptor 1
Tat	Transactivator of Transcription (HIV-1 protein)
TIPS	Treasury Inflation-Protected Securities
UDTA	Uniform Directed Trust Act
USEC	United States Enrichment Corporation

C Verification

I, CAUSTIN LEE McLAUGHLIN, declare under penalty of perjury pursuant to 28 U.S.C. ğ1746 that:

1. I am the author and data subject of the materials reproduced in this Master Filing;
2. I have read the foregoing Master Filing and know its contents;
3. The factual allegations and technical propositions are true and correct to the best of my knowledge, information, and belief;
4. The scientific, technical, and legal authorities cited are accurately represented to the extent cited as propositions;
5. I am proceeding *pro se* due to documented inability to secure retained counsel.

Executed on this _____ day of _____, 2026.

_____

CAUSTIN LEE MCLAUGHLIN
Pro Se
Baltimore, Maryland

END OF REVISED MASTER FILING